Yaming Zhang

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EDUCATION

Northeastern University, *Boston, MA*, **GPA: 4.0** *Dec 2023*
Master of Science in Electrical and Computer Engineering
Concentration: Computer Systems and Software

University of California Riverside, *Riverside, CA*, **GPA: 3.8** *Jun 2020*
Graduate Preparation Program in Computer Science(No Degree)

Xidian University, *Xi'an, China*, **GPA: 3.6** *Jun 2019*
Bachelor of Science in Computer Science

SKILLS

Languages: Java, Golang, Node.js, Scala
Frameworks/Libraries: Spring, gRPC, Spark, Kafka, Flink
Databases: MySql, Redis, MongoDB, Elasticsearch, ScyllaDB
DevOps: Kubernetes, Filebeat, Prometheus, Grafana

WORK EXPERIENCE

Electronic Arts (EA), *Redwood City, CA*
Software Engineer Intern *May 2023 – Present*

- Designed and implemented the Player Lifecycle Graph, a serialization model to manage players' states and unified messages in popular games such as FIFA and APEX.
- Developed an **Apache Flink** topology for **Apache Kafka** streaming trigger events execution, enabling efficient retrieval of lifecycles from **Redis**, reading player profiles from **ScyllaDB**, and executing the corresponding lifecycles.
- Utilized EADP's **Data Query Language (DQL)** to extract data from unstructured JSON events, evaluate boolean conditions, and execute action scripts.
- Built a web application with **React** and implemented APIs with **SpringBoot**, empowering campaign managers to draw lifecycle flowcharts and retrieve players' current states in games.

Tencent, *Beijing, China*
Software Engineer *Jun 2020 – Aug 2022*

- Developed multiple back-end servers using **gRPC** or **tRPC** with high-tolerance APIs, shipper for logs, monitoring system, asynchronous channels to **Apache Kafka**, etc to enhance reliability of Tencent services.
- Deployed Tencent SaaS services on **Tencent Cloud** via **Docker** containers and **Kubernetes** automated CI/CD construction to improve service scalability.
- Wrote customized codes in the secondary development of **Apache Superset** to create a dashboard displaying user usage data of Tencent services by preprocessing and visualizing data with Pandas and **Apache ECharts**.
- Analyzed the data of Tencent advertisements using **Apache Spark** to promote automatic advertising and personalized recommendation.

RESEARCH PROJECTS

OSMX *Feb 2021 – Present*

- Leveraged the power of **SparkSQL** to design and implement user-defined functions for geometry calculation, enabling the provision of diverse and flexible APIs that empower developers to generate and manipulate geometries with ease.
- Utilized the **Hadoop** file API to efficiently parse records from the **OpenStreetMap (OSM)** PBF file, enabling extraction of OSM entities for subsequent processing and analysis.
- Extracted nodes, POIs, roads and all objects using SparkSQL and categorized different datasets from all objects based on OSM tags, including parks, lakes and buildings, etc.
- Generated the satellite image of corresponding OSM Vector data using **Beast** Raptor spatial-join for spatial machine learning and AI activities.

Beast *Aug 2019 – Nov 2021*

- Developed the function of outputting and compressing the .kml file of geospatial datasets in Java to strengthen the I/O module, a basic module of UCR-STAR.
- Optimized functions of well-known text and well-known binary conversion in Geolite computational geometry library for the benchmark of the performance between **Esri**, **JTS** and Geolite API.
- Improved functions relevant to wkb format to enhance the compatibility of parsing speed with Geolite.
- Increased the granularity of Raptor join from pixels to tiles to improve the performance of spatial join for large data.

PUBLICATIONS

Samriddhi Singla, **Yaming Zhang**, Ahmed Eldawy (2022). OSMX: Spark-based Geospatial Data Extractor from OpenStreetMap and Satellite Images. International Conference on Advances in Geographic Information Systems.

Ahmed Eldawy, ..., **Yaming Zhang** (2021). Beast: Scalable Exploratory Analytics on Spatio-temporal Data. ACM International Conference on Information and Knowledge Management.

Yaming Zhang, Ahmed Eldawy (2020). Evaluating Computational Geometry Libraries for Big Spatial Data Exploration. International ACM SIGMOD Workshop on Managing and Mining Enriched Geo-Spatial Data.